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Document Code: WI-01-MIC-08	SHIGELLA		

1. OBJECTIVE

To determine the presence of *Shigella* organism in fish and fishery product samples.

2. SCOPE

This conventional method is applicable in the determination of *Shigella* organism in fish and fishery products samples.

3. RESPONSIBILITY

The analyst performing the test should ensure that the procedure is followed accordingly.

3.1 EQUIPMENT/LABORATORY APPARATUS

3.2 Alcohol lamps

3.3 Autoclave

3.4 Autoclave tape

3.5 Top loading Balance

3.6 Spatula

3.7 Forceps – stainless steel

3.8 Scissors – stainless steel

3.9 Graduated cylinder (100 ml and 500 ml, Pyrex)

3.10 Media bottles (500 ml and 1000 ml, Pyrex)

3.11 Sterile Pippetor

3.12 Sterile Petri dishes with cover (100x15mm, Pyrex)

3.13 Test Tubes with autoclavable caps (10 ml, 15 ml, and 20 ml, Pyrex)

3.14 Test tube racks

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- 3.15 Laboratory Blender/Stomacher
- 3.16 Incubators maintained at 35°C. Covered water bath, with circulating system to maintain temperature of $42.0 \pm 0.2^{\circ}\text{C}$ and $44.0 \pm 0.2^{\circ}\text{C}$.
- 3.17 Inoculating loops and needles
- 3.18 Biosafety Cabinet
- 3.19 Pipettes (must be top-calibrated) 10ml
- 3.20 Micro-pipette (must be top-calibrated) 0.1ml
- 3.21 Anaerobic jar with catalyst

4. MATERIALS AND CONSUMABLES

- 4.1 Stomacher bags
- 4.2 Tissue paper or paper towel
- 4.3 Alcohol(70% alcohol)
- 4.4 Glass Marking pencil
- 4.5 Surgical gloves
- 4.6 Surgical Mask

5. CULTURE MEDIA

- 5.1 Shigella broth with novobiocin
- 5.2 Trypticase Soy Yeast Extract broth
- 5.3 MacConkey Agar
- 5.4 Triple Sugar Iron
- 5.5 Urea Broth
- 5.6 Glucose Broth

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5.7 Lysine Decarboxylase Broth

5.8 Motility Agar

5.9 Tryptone (tryptophane) Broth

6. REAGENTS AND STAIN

6.1 Kovac's Reagent

6.2 Vogues Proskauer test reagents

6.3 1N Sodium Hydroxide

6.4 1N Hydrochloric Acid

6.5 Methyl Red Indicator

6.6 Novobiocin

6.7 Gram Stain Reagents

7 Pre- Analysis Procedure

7.1 Disinfect work area and equipment before analysis.

7.2 Complies with WI-02-02 Handling, Collections and Disposal of Hazardous Wastes.

7.3 Check if the samples are labelled with Job order numbers accompanied with analyst worksheet.

7.4 Handle samples individually to avoid mixing.

7.5 Prepare the media to be used that are appropriate to the analysis and commensurate to the number of samples to be analysed. All information regarding media preparation is placed in LF-W04-01, Reagent Preparation Logbook.

7.6 Properly label the media bottle, test tubes, dilution bottle and petri-dishes.

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8. PROCEDURE

8.1 Enrichment of *Shigella sonnei*

1. Aseptically weigh 25g of sample into Stomacher bag and add 225ml of sterile Shigella Broth to which novobiocin(0.5 µg/ml) is added.
2. Mix by placing the stomacher bag in the automatic bag mixer for 2mins. Hold suspension 10min at room temperature and shake periodically.
3. Pour supernatant into sterile 500ml Erlenmeyer flasks.
4. Adjust pH, if necessary, to 7.0 ± 0.2 with sterile 1N NaOH or 1N HCL.
5. Place flask in anaerobic jar with fresh catalyst, insert GasPak and activate by adding water.
6. Incubate jars in 44.0°C water bath for 20 hours.

Note: Enrichment of other *Shigella* species. Proceed as above(8.1) but use novobiocin at 3.0 µg/ml and incubate anaerobically in 42 °C water bath.

8.2 Plating on selective agar

1. After incubation at 44 °C water bath for 20h, agitate enrichment culture suspension and streak on a MacConkey agar plate.
2. Incubate at 35 °C for 20 hours.
3. Examine the plates for the presence *Shigella* colonies.
 - *Shigella* colonies are slightly pink and translucent, with or without rough edges.

7.7 Screening for confirmatory test

1. Inoculate suspicious colonies from MacConkey agar into glucose broth, TSI agar slant, lysine decarboxylase broth, motility agar and tryptone
2. Incubate at 35 °C for 48 hours but examine at 20 hours.
3. Discard all cultures showing motility, H₂S gas formation, lysine decarboxylation, and fermentation of sucrose or lactose.

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- With respect to indole formation, discard positive cultures from 44 °C enrichment.
- All suspicious isolates from 42°C enrichment may be either positive or negative should be retained.

Table 1. For confirmatory test

Positive to:	Tests	
	TSI	LIA
Shigella	Slant- Red Butt- yellow <i>without production of H₂S (blackening of agar)</i>	Slant- dark Violet Butt- yellow

Negative to:	Tests	
	TSI	LIA
Shigella	Slant- Yellow Butt- Yellow	Slant- light purple with yellow shadow Butt- light purple

8 BIO-CHEMICAL TEST FOR Shigella

8.1 Testing of Urease -negative cultures

- Lysine decarboxylase broth
- Inoculate broth with a small amount of growth from TSI / LIA slant

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suspicious for Shigella.

3. Replace cap tightly and incubate 48 ± 2 hours at 35°C .
4. Examine tubes at 24 hours interval.
5. Negative test is indicated by yellow reaction throughout the medium.
6. If medium appears discolored (neither purple nor yellow) add a few drops of 0.2 % bromocresol purple dye and re-read tubes reaction.

8.2 Pure Cultures

1. Urease test (conventional)
 - a. With sterile needle, inoculate growth from each presumed-positive TSI slant culture into tubes of urea broth.
 - b. Replace cap tightly and incubate 24 ± 2 hours at 35°C .
 - c. Examine tubes.
 - d. Positive test is indicated by purple-red on standing throughout the uninoculated tubes of urea broth.
2. Indole Test
 - a. With sterile needle, inoculate growth from each presumed-positive TSI slant culture into tubes of 5 ml tryptone broth.

Replace cap tightly and incubate 24 ± 2 hours at 35°C .
 - b. Add 0.2-0.3 ml Kovac's reagent. Read Results.
 - c. Negative test is lack of deep red color at surface of broth. Record intermediate shades of orange and pink.

8.3 Biochemical Tests (cultures that exhibit typical Shigella reactions)

1. MR-VP BROTH Test.

Inoculate medium with a small amount of growth from each unclassified TSI slant suspected to 5 ml MRVP broth which contain Shigella.

Incubate 48 ± 2 hours at 35°C .

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a. MR (methyl red) test

1. To 5 ml of 96 hours MR-VP broth, add 5-6 drops of methyl red.
2. Read results immediately.
3. Positive test indicated by diffuse red color in medium.
4. Negative test is distinct yellow color.

b. VP (Voges-Proskauer) test

1. Transfer 1 ml of 48 hours culture test tube and incubate remainder of MR-VP broth for an additional 48 hours at 35°C.
2. Add 0.6 ml α -naphthol and shake well.
3. Add 0.2 ml of 40% KOH solution and shake well. to intensify and speed reaction, add a few crystals of creatine.
4. Read results after 4 hours.
5. Positive test results are development of pink-to-ruby red color throughout the medium.

2. Simmons Citrate agar test.

Using sterile needle, inoculate growth from each presumed-positive TSI slant culture by streaking Simmons citrate agar slant and stabbing the butt.

- a. Replace cap tightly and incubate 96 ± 2 hours at 35 °C.
- b. Examine tubes. Read results.
- c. Positive test presence of growth, and is usually accompanied by color change from green to blue.
- d. Negative test no or very little growth and no color change.

9 INTERPRETATIONS

Interpret results of biochemical reactions as follows:

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Test for Shigella	Result
Urease	Negative(-)
Glucose Broth	-
Tryptone	-
MR	Positive (+)
Citrate	-
H ₂ S	-
Lysine Decarboxylase	-

10 REFERENCES

- Bacteriological Analytical Manual, Edition 8, Revision A, 1998. Chapter 3. Authors: Larry Maturin and James T. Peeler
- Food and Drug Administration. 1992. Bacteriological Analytical Manual, 7th ed. AOAC International, Arlington, VA, USA.

11 FORMS

LF W01-01, Reagent and Media Preparation Logbook

LF W01-MIC-03, Analyst Worksheet for Bacteriological Examination of Fish and Fishery Products

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